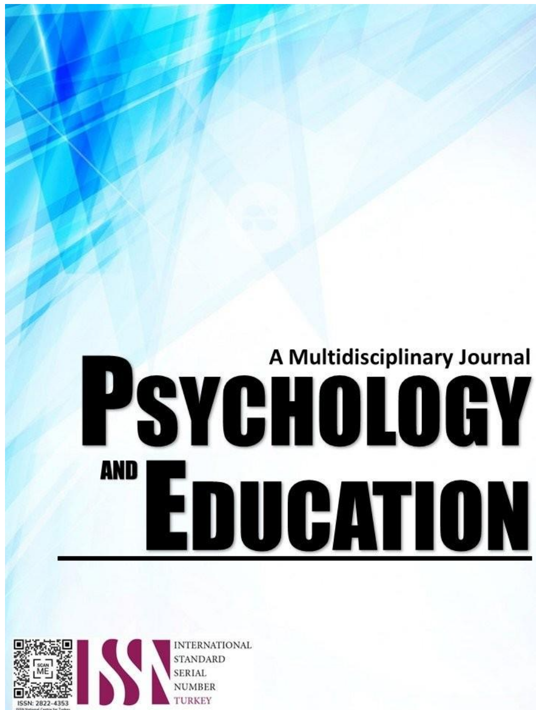
TEACHERS' LIVED EXPERIENCES WITH ICT INTEGRATION IN TEACHING AND LEARNING: A CASE STUDY



PSYCHOLOGY AND EDUCATION: A MULTIDISCIPLINARY JOURNAL

Volume: 40
Issue 9
Pages: 1160-1165
Document ID: 2025PEMJ3918
DOI: 10.70838/pemj.400902
Manuscript Accepted: 06-09-2025

Psych Educ, 2025, **40**(9): 1160-1165, Document ID:2025PEMJ3918, doi:10.70838/pemj.400902, ISSN 2822-4353
Research Article

Teachers' Lived Experiences with ICT Integration in Teaching and Learning: A Case Study

Sharlene Mae D. Sarueda,* Eingilbert C. Benolirao, Ariel L. Ramos
For affiliations and correspondence, see the last page.

Abstract

This qualitative case study explored the lived experiences, challenges, and coping mechanisms of teachers at Cansantic Elementary School in integrating Information and Communications Technology (ICT) into their teaching practices. While global studies highlighted systemic barriers such as inadequate infrastructure and policy gaps, this research focused on individual teacher perspectives to uncover the nuanced realities of ICT integration in a localized educational context. Guided by two primary research objectives—examining teachers' exposure to ICT utilization and their strategies for navigating integration challenges—the study identified two overarching themes: Digital Challenges and Enhanced Learning Support, and Social and Ethical Use and ICT Management. Findings revealed persistent barriers, including limited access to resources, outdated equipment, and insufficient ICT training. Despite these challenges, teachers demonstrated resilience by adopting innovative instructional methods, fostering peer collaboration, and promoting ethical digital practices. These adaptive strategies not only supported classroom instruction but also enhanced student engagement and learning outcomes. The study underscored the need for equitable access to digital tools, ongoing professional development, responsive administrative support, and robust technical infrastructure. By aligning institutional policies with the practical realities faced by educators, the research advocated for a more inclusive, responsive, and sustainable approach to ICT integration. Ultimately, this study contributed to the broader discourse on technology in education by amplifying teachers' voices and offering actionable insights for school leaders and policymakers seeking to bridge the digital divide.

Keywords: *information and communications technology (ICT), ICT integration*

Introduction

While much research on ICT integration emphasizes systemic challenges—such as limited resources, technical barriers, and policy gaps (Correos, 2014)—the individual experiences of teachers remain underexplored. Although the benefits of ICT in education are well-recognized, it is the teachers, as frontliners of classroom innovation, who must navigate the daily realities of digital transformation. Understanding their perspectives offers valuable insight into the practical, personal, and contextual factors that shape the implementation of ICT in schools.

Global studies suggest that infrastructure and training alone do not guarantee effective ICT integration. In Europe and Asia, for example, teachers' autonomy, cultural context, and alignment with curriculum standards influence adoption (Schmid et al., 2020; Chong & Gan, 2022), while African and Latin American findings emphasize localized challenges and the need for context-specific support (Osakwe, 2021; Martínez & Rivera, 2023). These findings underscore the importance of investigating how teachers' lived experiences impact technology use in classrooms, especially in varied educational settings.

However, in the Philippine context, particularly in public elementary schools, teachers still face uneven access to ICT resources, inconsistent training opportunities, and limited institutional support. At Cansantic Elementary School, these challenges are evident: outdated or insufficient devices, shared internet connections, and a lack of ICT-related professional development hinder teachers from fully integrating digital tools into instruction. This study, therefore, seeks to explore the lived experiences, barriers, and coping strategies of these teachers, contributing a localized perspective to the broader discourse on ICT in education and identifying ways to better support technology integration at the grassroots level.

Research Questions

1. What are the participants' experiences and exposures in utilizing ICT while performing their duties?
2. How do the participants navigate the use of ICT in the course of performing their duties?

Literature Review

The integration of Information and Communication Technology (ICT) in education has been extensively studied, emphasizing its transformative impact on teaching, learning, and leadership. Mezirow's (2003) Transformative Learning Theory highlights how teachers and learners adapt their worldviews through reflection and reevaluation, enabling them to embrace innovative practices like ICT. However, successful integration depends heavily on leadership, as noted by Oznacar and Dericioğlu (2016), who stress that school administrators must possess motivation, creativity, and skills to address challenges such as inadequate infrastructure and limited technical support. Effective technology leadership, as emphasized by Raman and Thannimalai (2019), involves implementing policies that foster ICT use, making it a crucial factor in achieving educational goals.

Teachers' professional development is also vital, as Katz's Skills Theory underscores the importance of acquired skills in ensuring teacher effectiveness. Studies by Correos (2014) and Tuazon (2019) reveal that while many teachers have basic ICT skills, gaps remain in advanced technical operations, highlighting the need for sustained training. Programs like the Department of Education Computerization Program (DCP) aim to bridge these gaps by providing technological resources to enhance teaching and learning processes. However, the success of such programs often relies on the initiative of teachers and administrators to maximize their potential.

Learners' ICT skills play an equally critical role, as noted by Okan (2016), who highlights the necessity of digital literacy for academic and career success. With students often surpassing their teachers in ICT proficiency, educators must continually update their skills to effectively lead ICT-integrated classrooms. Global frameworks such as the TPACK model, the European Commission's DigCompEdu, and UNESCO's ICT Competency Framework provide comprehensive guidelines for developing teachers' digital competencies. These frameworks integrate technical, pedagogical, and organizational dimensions, ensuring meaningful learning through ICT.

The COVID-19 pandemic further underscored the importance of ICT literacy, as the rapid shift to online learning revealed gaps in educators' readiness but also created opportunities for systemic improvements. In the Philippine context, initiatives like the Basic Education Learning Continuity Plan (DepEd Order No. 12, series of 2020) and the constitutional mandate for quality education aim to create ICT-rich environments that support innovative teaching methods and transparent governance. Despite challenges, the integration of ICT in education requires transformative leadership, continuous professional development, and collaboration among stakeholders to ensure quality education for all.

ICT is an element that is yet to be exploited at the same level in educational settings as in other social and essential areas, and therefore its scope and growth are still quite broad. There is still a long way to go to achieve adequate initial training for teachers, as well as the professional development that this may entail for them. The mere fact of the existence of ICT does not change the educational process, instead, it is necessary to pay attention to the level, and above all the manner, of its implementation and teacher training for its proper use. To establish this strategy for the improvement and acquisition of digital skills by teachers, it is essential to know their lived experiences on these skills.

Methodology

This study employed a single case study design to explore the lived experiences of teachers at Cansantic Elementary School in integrating Information and Communication Technology (ICT) into their teaching practices. A case study is appropriate for this inquiry because it enables an in-depth investigation of a real-life situation within its natural context, allowing for a holistic understanding of complex experiences (Yin, 2018). By focusing on one school, the study captures specific insights that broader quantitative methods might overlook.

The research was conducted at Cansantic Elementary School, a small public school located in Sibonga, Cebu, offering Kindergarten to Grade VI education. The school was selected due to its relevance and accessibility, and because its size allowed for comprehensive participant involvement.

All nine teachers at the school participated in the study, using a total population sampling approach. This purposive technique was chosen to ensure that each teacher, regardless of subject area or grade level, had the opportunity to share their experience with ICT. Their diverse teaching roles enriched the data and made the findings more reflective of the school's overall ICT integration efforts.

Data were collected using a semi-structured interview guide designed to elicit detailed responses on teachers' ICT-related experiences, challenges, and coping mechanisms. The guide aligned with the study's objectives and allowed for open-ended, reflective conversations.

For data analysis, the study followed Miles and Huberman's (1994) framework, which includes data reduction, data display, and conclusion drawing. To enhance analytical rigor, Saldaña's (2021) coding methods were also employed, offering a systematic and layered approach to theme development. Together, these frameworks enabled a rich, accurate portrayal of participants' experiences.

The study's trustworthiness was ensured through Lincoln and Guba's (1985) criteria—credibility, transferability, dependability, and confirmability. Additionally, Tracy's (2010) qualitative quality framework provided further depth, emphasizing sincerity, ethical integrity, and data richness.

Ethical considerations were grounded in the principles of Kvale and Brinkmann (2015), which emphasize informed consent, confidentiality, minimization of harm, and researcher transparency. All participants were fully informed about the study's purpose and voluntarily agreed to participate, with assurances of anonymity and data protection.

Results and Discussion

Exposures of the Participants in the Context of ICT Utilization

The findings present varied responses from the teachers of Cansantic Elementary School, which were derived using Miles and Huberman's (1994) Thematic Framework. Data were gathered through interviews that explored the teachers' lived experiences, the

Psych Educ, 2025, 40(9): 1160-1165, Document ID:2025PEMJ3918, doi:10.70838/pemj.400902, ISSN 2822-4353
Research Article

challenges they faced, and their coping mechanisms in using information and communications technology (ICT) in teaching. These insights offer a grounded understanding of how ICT is being integrated into classroom practices within this specific educational setting.

Table 1. *Exposures of the Participants in the Context of ICT Utilization*

Themes	*Categories*
Digital Challenges	• Lack of Digital Skills and Literacy • Lack of access to the necessary infrastructure to support ICT as well as the gadgets and other instruments and legit software • Adaptability and updating oneself to the latest trends • Vigilant against fraud and fake accounts
Enhanced Learning Support	• Visually appealing to the learners • Teaching and learning efficiency • Providing technical assistance

The participants shared their experiences regarding the use of ICT in fulfilling their teaching responsibilities. These narratives were interpreted directly from the investigator's perspective using a thematic analysis approach. As shown in Table 1, the teachers' experiences at Cansantic Elementary School revealed two major themes: digital challenges and enhanced learning support, each supported by specific subcategories.

Digital Challenges

Participants reported various challenges in ICT utilization that negatively impact the teaching and learning process. Becta (2004) noted that limited access to ICT resources is not solely due to the absence of hardware or software but may also stem from poor resource organization, substandard equipment, or a lack of personal access for teachers. Similarly, Osborne and Hennessy (2003) emphasized that restricted access to technological tools diminishes teachers' motivation to integrate ICT into classroom instruction.

This difficulty is seen and is observed in the response of Participant No. 5 below:

"My computer just suddenly stopped working or sometimes my Microsoft applications expired or the system detects its ingenuity and there is nothing I can do about it but to call our School ICT Coordinator for help because I still have a class to teach as well as IMS to print and all other reports to comply. It isn't easy on my part because of deadlines are already set.

Many teachers lack the skills to effectively use ICT in the teaching-learning process due to insufficient training opportunities. This lack of training has contributed to reluctance in adopting technology within the classroom. As emphasized by Beggs (2000), inadequate training remains one of the top three barriers to ICT integration in education. Becta (2004) also noted that effective training requires multiple components, including sufficient time, pedagogical orientation, technical skills, and integration of ICT in initial teacher preparation. Providing pedagogical—not just technical—training is crucial to help teachers use ICT meaningfully to support student learning.

This difficulty is seen and is observed in the response of Participant No. 1 below:

"I call the assistance of our ICT Coordinator because she's the only person here who is like an expert in dealing with ICT problems, whenever I cannot open the applications in our computer or when I cannot send in my GCash. We've had trainings before but it's too vague."

Several studies have outlined key reasons for limited access to technology in schools. Sicilia (2005) highlighted teachers' difficulties in accessing computers, often due to booking conflicts or shared resources that hindered sustained use during instructional projects. Teachers reported challenges such as insufficient computers, limited peripherals, inadequate software licenses, and restricted internet access as major barriers to ICT implementation. Balanskat et al. (2006) emphasized that access alone does not ensure effective ICT integration, pointing to issues like substandard hardware, lack of relevant educational software, and limited accessibility. Newhouse (2002) added that mismatched hardware and software choices often complicate classroom use. Similarly, Cox et al. (1999) found that most teachers cited insufficient resources and limited time to explore software as significant barriers. Other mid-level obstacles included teacher reluctance, low motivation, financial constraints, administrative delays, limited technical support, and insufficient time for engagement.Participant No. 8 shared her sentiments that she needs to get a loan to purchase a gadget so she will be able to deliver her tasks every day.

Mangutang ko mam para makapalit ko sa akong kinahanglanon sa eskwelahan sama sa laptop, dili man gane ni katong mga I series kay mahal kayo. Hasta ning cellphone mam inutang sad ni kay gamit man gyud kayo ni sa atong pagpanarbaho. Unsaon ang Deped maoy atong saligan hatagan man tag kung dili gubaon hayan pirte pung loadinga nga item. Maningkamot nalang tag atua ani para naa tay magamit sa klase.

Given these persistent challenges, it is crucial to address the barriers that hinder effective ICT utilization to improve teaching and learning outcomes (UNESCO, 2021). Without resolving issues such as inadequate training, limited infrastructure, and low teacher motivation, technology integration remains superficial. As the discussion now turns to the second theme—enhanced learning support—

we examine how targeted interventions and strategic resource allocation can empower both teachers and students to leverage ICT more meaningfully in the classroom.Enhanced Learning Support

An important role of ICT inside a school is that of providing a new framework that can foster a revision and an improvement of teaching and learning practices. Teachers will need curricula-related content and clear strategies and examples to effectively use these materials in the classroom. They will find that students can become highly motivated toward learning (and even toward regularly attending school) if technology makes up part of their classroom experience. Teachers can fruitfully use this positive attitude to explore new learning strategies in which students can be more actively involved in learning and as well as develop their creativity, as opposed to being simply passive information receivers. Creativity is very important to develop from an early age. Creativity allows humans to improve their quality of life (Munandar, 1999). Considering the importance of ICT in facilitating students to carry out their creativity in their learning activities, the use of ICT in learning activities must be adjusted to the stated learning objectives. The presence of ICT in the learning process helps students better understand the subject matter being studied. But in the field, there are still many teachers who find it difficult to utilize ICT in learning so that the desired learning objectives can be more easily achieved. In the end, the use of ICT will support the effectiveness efficiency, and attractiveness of organizing learning activities as stated in the following response by the participant no. 3

"Nindot man jud siya mam, it is very effective and efficient in the delivery of lessons since it is visually appealing to the eyes of the learners. It captures pupils' attention, provides an innovative vision for the school, and enhances interaction and rapport. But sometimes we'll just rely on what is provided on the internet because it's very time consuming if we are going to create something as interactive as what the internet can provide for us plus we still need to watch tutorials on how to do this and that aside from the fact that we need to have 8 lesson preparations every day plus our ancillaries and other side tasks to finish for the day. Sometimes we'll pay in Facebook so we'll have the PowerPoint lessons and DLPs that we can use for our pupils.

Ways of Navigating with ICT While Performing Functions

The succeeding table summarizes the different categories picked out from the actual responses to form the emerging themes relevant to various ways that successful teachers have in navigating while with ICT performing functions.

Table 2. *Ways of Navigating with ICT*

Themes	*Categories*
Social and Ethical use of ICT	• Apply personal security protocols • Apply digital information security practices • Recognize intellectual property
ICT Management and Operations	• Selecting and using hardware and software • Generates ideas, plans and processes and solutions to challenges and learning area tasks • Collaborate, share and exchange ideas during in-service training for teachers

Social and Ethical use of ICT

As teachers integrate ICT into their daily classroom activities, they are trained to properly acknowledge the sources of their learning materials and protect their digital footprint to avoid scams and security breaches. They adopt appropriate practices to respect intellectual property rights, securely store digital information, and follow protocols to safely create, communicate, and share information. Teachers also develop an awareness of the societal benefits and consequences of ICT use by individuals and communities. ICT integration has been discussed among educators for over three decades (Lowther, Strahl, Inan, & Ross, 2008), with thousands of studies published recommending strategies to support meaningful, student-centered technology use (Becker & Riel, 1999). Many of these focus on overcoming barriers encountered by schools and teachers, a challenge echoed by participant no. 6 in this study.

I always double-check the accuracy of my sources before giving it to my students to ensure that they are getting the correct information about the lesson. I also verify the sources to see to it that it is correct especially that in the internet there are a lot of misleading information on a certain topic. We need to be vigilant because nowadays grabe na kaayo ang mga fake news peddler and isa pa sad one thing I learned sa among ICT Coor is to update our passwords kanunay to avoid phishing scams etc. I also coordinated with our ICT Coordinator if ever we have concerns and confusions since she is very willing to assist us in our concerns but sometimes the real deal in the classroom setting is the lack of resources for our children to use which can hinder us from teaching the competency to our learners.

Ertmer (1999) identified two main types of barriers affecting teachers' use of technology in the classroom. First-order barriers are external factors such as lack of resources, inadequate training, and insufficient support. In contrast, second-order barriers are internal, including teachers' confidence, beliefs about student learning, and perceptions of technology's value in education. Although first-order barriers have long been recognized as significant obstacles to technology integration (O'Mahony, 2003; Pelgrum, 2001), addressing second-order barriers is equally important for successful implementation.

I use YouTube when I want to know how certain things are made especially during an experiment activity or output making. During a simple baking class, I use YouTube to see how things are made exactly to avoid mistakes. The pupils or the learners were actively engaged with learning because they could see what exactly to do with the help of the tutorials on the internet. Problems arise when the internet can no longer cater to the demands of video viewing. It caused slow browsing that can delay the lesson delivery.

In 2007, Hew and Brush analyzed barriers to technology integration documented over the previous decade. They identified six categories: four first-order barriers (resources, institution, subject culture, and assessment) and two second-order barriers (teacher attitudes and beliefs, and knowledge and skills). From their review of 48 empirical studies, the three most frequently cited obstacles were resources (40%), teachers' knowledge and skills (23%), and teachers' attitudes and beliefs (13%). By the early 2000s, increased access to technology resources (National Center for Education Statistics, 2006) helped reduce the impact of first-order barriers. Consequently, research shifted focus toward how teachers' pedagogical beliefs influence meaningful technology use, especially in supporting student-centered learning (Dexter & Anderson, 2002; Ertmer, 2005; Judson, 2006). Means and Olson (year) described student-centered learning as using technology to engage students in collaborative, authentic, and challenging tasks that foster inquiry and joint investigation. McCain (2005) emphasized that the key educational challenge is not technology use itself but developing students' thinking skills to leverage technology for problem-solving and meaningful work. This approach requires placing technology in students' hands to promote communication, collaboration, and problem-solving as professionals do. The following statements illustrate some participants' struggles with ICT integration in their classrooms.

Participant No. 4 shared that,

"During the in-service training, we are given various applications that we can utilize to make our classroom discussions interesting. Some of them are online quizzes where pupils can interact online. But the sad part is that not all of my pupils have the device. But overall, it made my teaching instruction easy, colorful, and interesting."

In summary, Table 2 presents the responses of the teachers at Cansantic Elementary School on how they navigate ICT utilization by employing various strategies to enhance productivity and efficiency. They use LAC sessions as valuable opportunities to generate, gather, and share ideas and best practices for maximizing ICT in teaching, protecting digital data, and respecting intellectual property (Becta, 2004). The data clearly show that collaboration and sharing expertise serve as effective strategies for managing ICT integration in a rapidly evolving educational landscape where technology plays a crucial role (Johnson et al., 2016).

Conclusions

The findings of this study revealed two main realities in the ICT integration practices of Cansantic Elementary School teachers: significant digital challenges and innovative strategies that enhance learning support and ethical ICT use. The digital challenges included limited access to functioning hardware, outdated equipment, financial constraints, shared resources, and insufficient training. These obstacles disrupted workflow and reflected barriers identified in previous research related to resources, organizational inefficiencies, and skill gaps.

Despite these challenges, teachers demonstrated resilience by using ICT to boost student engagement through interactive content and online platforms such as YouTube. They actively participated in Learning Action Cell (LAC) sessions to share best practices and employed ethical digital behaviors to safeguard information. This highlights their commitment to fostering responsible technology use and digital citizenship in their classrooms.

Systemic issues like slow internet and administrative burdens also hindered effective ICT integration. However, the teachers' collaborative spirit and proactive efforts suggest that with adequate support, they can overcome these barriers to create a more inclusive and technology-driven learning environment.

To improve ICT integration, the study recommends ensuring equitable access to updated hardware and reliable internet, providing sustained professional development, and streamlining administrative tasks. Institutionalizing collaborative platforms like LAC sessions and enhancing technical support systems will further empower teachers. Lastly, establishing a monitoring system to evaluate ICT's impact will help refine practices and promote successful models for other schools.

References

Agarwal, R., & Karahanna, E. (2000). Time flies when you're having fun: Cognitive absorption and beliefs about information technology usage. MIS Quarterly, 24(4), 665–694.

Ajzen, I., Brown, T. C., & Carvajal, F. (2004). Explaining the discrepancy between intentions and actions: The case of hypothetical bias in contingent valuation. Personality and Social Psychology Bulletin, 30(9), 1108–1121.

Bassellier, G., Benbasat, I., & Reich, B. H. (2003). The influence of business managers' IT competence on championing IT. Information Systems Research, 14(4), 317–336.

Bassellier, G., Reich, B. H., & Benbasat, I. (2001). Information technology competence of business managers: A definition and research model. Journal of Management Information Systems, 17(4), 159–182.

Baylor, A. L., & Ritchie, D. (2002). What factors facilitate teacher skill, teacher morale, and perceived student learning in technology-using classrooms? Computers & Education, 39(4), 395–414.

Rubenstein, D. (2007). Standish group report: There's less development chaos today. SD times on the Web, http://www.sdtimes.com/content/article.aspx?ArticleID=30247 Accessed 12.05.08.

Schiller, J. (2003). Working with ICT: Perceptions of Australian principals. Journal of Educational Administration, 41(2), 171–185.

Sheeran, P. (2002). Intention-behaviour relations: A conceptual and empirical review. In W. Stroebe & M. Hewstone (Eds.), European review of social psychology (12). Chichester: Wiley.

Slowinski, J. (2003). Becoming a technologically savvy administrator. Teacher Librarian, 30(5), 25–29.

Tearle, P. (2003). ICT implementation: What makes the difference? British Journal of Educational Technology, 34(5), 567–583.

Telem, M. (2001). Computerization of school administration: Impact on the principal's role – A case study. Computers & Education, 37, 345–362.

Testerman, J. C., Flowers, C. P., & Algozzine, B. (2001). Basic technological competencies of educational administrators. Contemporary Education, 72(2), 58–63.

UNESCO. (2015). Education 2030. Incheon Declaration and Framework for Action for the Implementation of Sustainable Development Goal 4, page 8, at: https://en.unesco.org/education2030-sdg4

UNESCO. (2015). Qingdao Declaration, page 2, at: http://www.unesco.org/new/fileadmin/MULTIMEDIA/HQ/ED/pdf/Qingdao_Declaration.pdf

WHO. (2011). World Disability Report, at http://www.who.int/disabilities/world_report/2011/en/

UNESCO. (2016). Learning for All: Guidelines on the Inclusion of Learners with Disabilities in Open and Distance Learning, at: http://unesdoc.unesco.org/images/0024/002443/244355e.pdf 8 See UNESCO. (2015). Education 2030. Incheon Declaration and Framework for Action for the Implementation of Sustainable Development Goal 4, page 28, at: https://en.unesco.org/education2030-sdg4

Simon Buckingham Shum. (2012). UNESCO IITE Policy Brief: Learning Analytics, at: http://iite.unesco.org/pics/publications/en/files/3214711.pdf

Affiliations and Corresponding Information

Sharlene Mae D. Sarueda
Cebu Technological University - Argao Campus – Philippines

Eingilbert C. Benolirao
Cebu Technological University - Argao Campus – Philippines

Ariel L. Ramos
Cebu Technological University - Argao Campus – Philippines